

MongoDB CRUD Operations

1. Create Database

```
[test> use FreelanceMarketplace
switched to db FreelanceMarketplace
```

2. Create Collections and Insert Documents

`insertOne()`

Used to insert a **single document** into a MongoDB collection.

`insertMany()`

Used to insert **multiple documents at once** into a MongoDB collection.

Freelancers Collection

```
FreelanceMarketplace> db.Freelancers.insertOne({
|   freelancer_id: 101,
|   name: "Aarav Sharma",
|   skill: "Web Development",
|   experience: 3,
|   hourly_rate: 35,
|   rating: 4.7,
|   location: "Bangalore"
| })
[{
  acknowledged: true,
  insertedId: ObjectId('6a7c0cbdeb9e6b1227837edd')
}]
```

```

FreelanceMarketplace> db.Freelancers.insertMany([
  {
    freelancer_id: 102,
    name: "Priya Mehta",
    skill: "Graphic Design",
    experience: 4,
    hourly_rate: 30,
    rating: 4.8,
    location: "Delhi"
  },
  {
    freelancer_id: 103,
    name: "Rahul Verma",
    skill: "Data Science",
    experience: 5,
    hourly_rate: 45,
    rating: 4.9,
    location: "Mumbai"
  },
  {
    freelancer_id: 104,
    name: "Ananya Singh",
    skill: "Content Writing",
    experience: 2,
    hourly_rate: 20,
    rating: 4.5,
    location: "Pune"
  },
  {
    freelancer_id: 105,
    name: "Karan Patel",
    skill: "App Development",
    experience: 6,
    hourly_rate: 50,
    rating: 4.6,
    location: "Ahmedabad"
  }
])
[
  {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6a7c0cd3eb9e6b1227837ede'),
      '1': ObjectId('6a7c0cd3eb9e6b1227837edf'),
      '2': ObjectId('6a7c0cd3eb9e6b1227837ee0'),
      '3': ObjectId('6a7c0cd3eb9e6b1227837ee1')
    }
  }
]

```

Clients Collection

```

FreelanceMarketplace> db.Clients.insertMany([
  {
    client_id: 201,
    name: "TechNova Solutions",
    industry: "Technology",
    budget: 5000,
    location: "Bangalore"
  },
  {
    client_id: 202,
    name: "CreativeHub",
    industry: "Marketing",
    budget: 3000,
    location: "Delhi"
  },
  {
    client_id: 203,
    name: "DataWorks",
    industry: "Analytics",
    budget: 7000,
    location: "Mumbai"
  },
  {
    client_id: 204,
    name: "EduSmart",
    industry: "Education",
    budget: 4000,
    location: "Pune"
  },
  {
    client_id: 205,
    name: "HealthPlus",
    industry: "Healthcare",
    budget: 6000,
    location: "Hyderabad"
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a7c0ce0eb9e6b1227837ee2'),
    '1': ObjectId('6a7c0ce0eb9e6b1227837ee3'),
    '2': ObjectId('6a7c0ce0eb9e6b1227837ee4'),
    '3': ObjectId('6a7c0ce0eb9e6b1227837ee5'),
    '4': ObjectId('6a7c0ce0eb9e6b1227837ee6')
  }
}

```

Projects Collection

```

FreelanceMarketplace> db.Projects.insertMany([
  {
    project_id: 301,
    title: "E-Commerce Website",
    client_id: 201,
    freelancer_id: 101,
    category: "Web Development",
    budget: 4500,
    status: "Completed",
    hours: 120
  },
  {
    project_id: 302,
    title: "Brand Logo Design",
    client_id: 202,
    freelancer_id: 102,
    category: "Graphic Design",
    budget: 1800,
    status: "In Progress",
    hours: 40
  },
  {
    project_id: 303,
    title: "Sales Prediction Model",
    client_id: 203,
    freelancer_id: 103,
    category: "Data Science",
    budget: 6000,
    status: "Completed",
    hours: 150
  },
  {
    project_id: 304,
    title: "Educational Blog",
    client_id: 204,
    freelancer_id: 104,
    category: "Content Writing",
    budget: 2500,
    status: "In Progress",
    hours: 60
  },
  {
    project_id: 305,
    title: "Healthcare Mobile App",
    client_id: 205,
    freelancer_id: 105,
    category: "App Development",
    budget: 7500,
    status: "Pending",
    hours: 200
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a7c0e33eb9e6b1227837ee7'),
    '1': ObjectId('6a7c0e33eb9e6b1227837ee8'),
    '2': ObjectId('6a7c0e33eb9e6b1227837ee9'),
    '3': ObjectId('6a7c0e33eb9e6b1227837eea'),
    '4': ObjectId('6a7c0e33eb9e6b1227837eeb')
  }
}

```

3. Read Operations

find()

Used to retrieve **multiple documents** from a collection. It can also be used with filters to retrieve specific records.

findOne()

Used to retrieve **one document** that matches the given

```
[FreelanceMarketplace> db.Freelancers.find()
[
  {
    _id: ObjectId('6a7c0cbdeb9e6b1227837edd'),
    freelancer_id: 101,
    name: 'Aarav Sharma',
    skill: 'Web Development',
    experience: 3,
    hourly_rate: 35,
    rating: 4.7,
    location: 'Bangalore'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ede'),
    freelancer_id: 102,
    name: 'Priya Mehta',
    skill: 'Graphic Design',
    experience: 4,
    hourly_rate: 30,
    rating: 4.8,
    location: 'Delhi'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837edf'),
    freelancer_id: 103,
    name: 'Rahul Verma',
    skill: 'Data Science',
    experience: 5,
    hourly_rate: 45,
    rating: 4.9,
    location: 'Mumbai'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ee0'),
    freelancer_id: 104,
    name: 'Ananya Singh',
    skill: 'Content Writing',
    experience: 2,
    hourly_rate: 20,
    rating: 4.5,
    location: 'Pune'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ee1'),
    freelancer_id: 105,
    name: 'Karan Patel',
    skill: 'App Development',
    experience: 6,
    hourly_rate: 50,
    rating: 4.6,
    location: 'Ahmedabad'
  }
]
```

condition.]

```
[FreelanceMarketplace> db.Freelancers.findOne({ freelancer_id: 101 })
{
  _id: ObjectId('6a7c0cbdeb9e6b1227837edd'),
  freelancer_id: 101,
  name: 'Aarav Sharma',
  skill: 'Web Development',
  experience: 3,
  hourly_rate: 35,
  rating: 4.7,
  location: 'Bangalore'
}
```

Filter


```

FreelanceMarketplace> db.Freelancers.find({ skill: "Data Science" })
[
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837edf'),
    freelancer_id: 103,
    name: 'Rahul Verma',
    skill: 'Data Science',
    experience: 5,
    hourly_rate: 45,
    rating: 4.9,
    location: 'Mumbai'
  }
]

FreelanceMarketplace> db.Projects.find({ status: "Completed" })
[
  {
    _id: ObjectId('6a7c0e33eb9e6b1227837ee7'),
    project_id: 301,
    title: 'E-Commerce Website',
    client_id: 201,
    freelancer_id: 101,
    category: 'Web Development',
    budget: 4500,
    status: 'Completed',
    hours: 120
  },
  {
    _id: ObjectId('6a7c0e33eb9e6b1227837ee9'),
    project_id: 303,
    title: 'Sales Prediction Model',
    client_id: 203,
    freelancer_id: 103,
    category: 'Data Science',
    budget: 6000,
    status: 'Completed',
    hours: 150
  }
]

```

4. Logical Operators

\$and

Returns documents only when all specified conditions are true.

```

FreelanceMarketplace> db.Freelancers.find({
|   $and: [
|     { experience: { $gt: 3 } },
|     { rating: { $gt: 4.6 } }
|   ]
| })
[
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ede'),
    freelancer_id: 102,
    name: 'Priya Mehta',
    skill: 'Graphic Design',
    experience: 4,
    hourly_rate: 30,
    rating: 4.8,
    location: 'Delhi'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837edf'),
    freelancer_id: 103,
    name: 'Rahul Verma',
    skill: 'Data Science',
    experience: 5,
    hourly_rate: 45,
    rating: 4.9,
    location: 'Mumbai'
  }
]

```

\$or

Returns documents when at least one of the specified conditions is

```

FreelanceMarketplace> db.Freelancers.find({
|   $or: [
|       { skill: "Data Science" },
|       { skill: "App Development" }
|   ]
| })
[
|   {
|     _id: ObjectId('6a7c0cd3eb9e6b1227837edf'),
|     freelancer_id: 103,
|     name: 'Rahul Verma',
|     skill: 'Data Science',
|     experience: 5,
|     hourly_rate: 45,
|     rating: 4.9,
|     location: 'Mumbai'
|   },
|   {
|     _id: ObjectId('6a7c0cd3eb9e6b1227837ee1'),
|     freelancer_id: 105,
|     name: 'Karan Patel',
|     skill: 'App Development',
|     experience: 6,
|     hourly_rate: 50,
|     rating: 4.6,
|     location: 'Ahmedabad'
|   }
| ]
true.

```

\$not

Returns documents where the specified condition is not true.


```

FreelanceMarketplace> db.Freelancers.find({
|   experience: { $not: { $gt: 4 } }
| })
[
  {
    _id: ObjectId('6a7c0cbdeb9e6b1227837edd'),
    freelancer_id: 101,
    name: 'Aarav Sharma',
    skill: 'Web Development',
    experience: 3,
    hourly_rate: 35,
    rating: 4.7,
    location: 'Bangalore'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ede'),
    freelancer_id: 102,
    name: 'Priya Mehta',
    skill: 'Graphic Design',
    experience: 4,
    hourly_rate: 30,
    rating: 4.8,
    location: 'Delhi'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ee0'),
    freelancer_id: 104,
    name: 'Ananya Singh',
    skill: 'Content Writing',
    experience: 2,
    hourly_rate: 20,
    rating: 4.5,
    location: 'Pune'
  }
]

```

\$nor

Returns documents that do not satisfy any of the given conditions.

```

FreelanceMarketplace> db.Freelancers.find({
|   $nor: [
|     { location: "Delhi" },
|     { location: "Mumbai" }
|   ]
| })
[
  {
    _id: ObjectId('6a7c0cbdeb9e6b1227837edd'),
    freelancer_id: 101,
    name: 'Aarav Sharma',
    skill: 'Web Development',
    experience: 3,
    hourly_rate: 35,
    rating: 4.7,
    location: 'Bangalore'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ee0'),
    freelancer_id: 104,
    name: 'Ananya Singh',
    skill: 'Content Writing',
    experience: 2,
    hourly_rate: 20,
    rating: 4.5,
    location: 'Pune'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ee1'),
    freelancer_id: 105,
    name: 'Karan Patel',
    skill: 'App Development',
    experience: 6,
    hourly_rate: 50,
    rating: 4.6,
    location: 'Ahmedabad'
  }
]

```

5. Relational Operators

\$gt — Greater Than

Selects documents where a field value is greater than the specified value.

```
[FreelanceMarketplace> db.Freelancers.find({ hourly_rate: { $gt: 30 } })
[
  {
    _id: ObjectId('6a7c0cbdeb9e6b1227837edd'),
    freelancer_id: 101,
    name: 'Aarav Sharma',
    skill: 'Web Development',
    experience: 3,
    hourly_rate: 35,
    rating: 4.7,
    location: 'Bangalore'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837edf'),
    freelancer_id: 103,
    name: 'Rahul Verma',
    skill: 'Data Science',
    experience: 5,
    hourly_rate: 45,
    rating: 4.9,
    location: 'Mumbai'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ee1'),
    freelancer_id: 105,
    name: 'Karan Patel',
    skill: 'App Development',
    experience: 6,
    hourly_rate: 50,
    rating: 4.6,
    location: 'Ahmedabad'
  }
]
```

\$lt — Less Than

Selects documents where a field value is less than the specified value.

```
[FreelanceMarketplace> db.Freelancers.find({ hourly_rate: { $lt: 40 } })
[
  {
    _id: ObjectId('6a7c0cbdeb9e6b1227837edd'),
    freelancer_id: 101,
    name: 'Aarav Sharma',
    skill: 'Web Development',
    experience: 3,
    hourly_rate: 35,
    rating: 4.7,
    location: 'Bangalore'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ede'),
    freelancer_id: 102,
    name: 'Priya Mehta',
    skill: 'Graphic Design',
    experience: 4,
    hourly_rate: 30,
    rating: 4.8,
    location: 'Delhi'
  },
  {
    _id: ObjectId('6a7c0cd3eb9e6b1227837ee0'),
    freelancer_id: 104,
    name: 'Ananya Singh',
    skill: 'Content Writing',
    experience: 2,
    hourly_rate: 20,
    rating: 4.5,
    location: 'Pune'
  }
]
```

\$eq — Equal To

Selects documents where a field value is equal to the specified value.



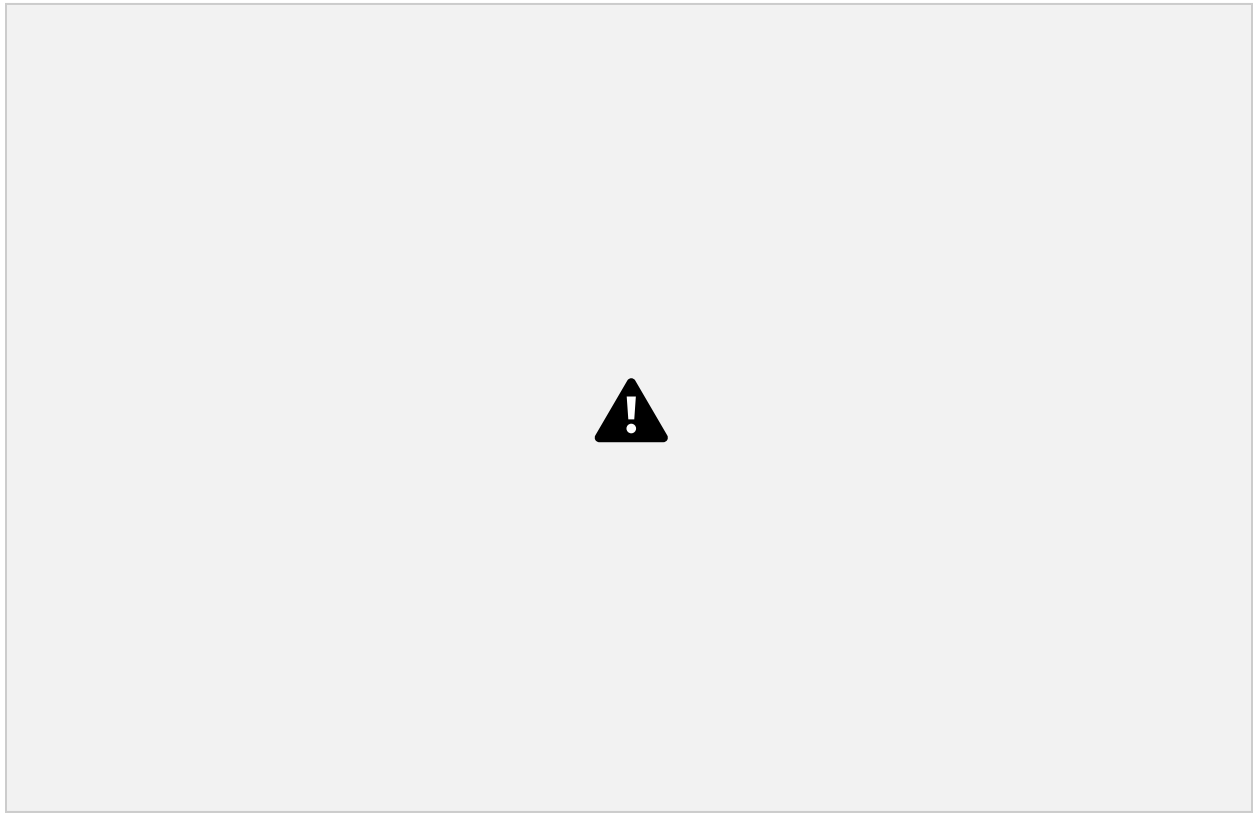
\$ne — Not Equal To

Selects documents where a field value is not equal to the specified value.



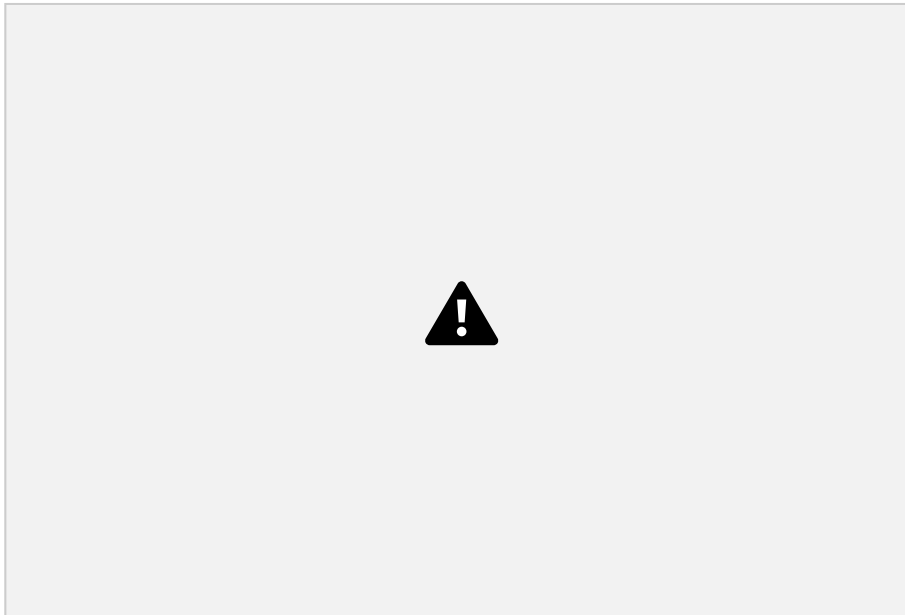
\$gte — Greater Than or Equal To

Selects documents where a field value is greater than or equal to the specified value.



\$lte — Less Than or Equal To

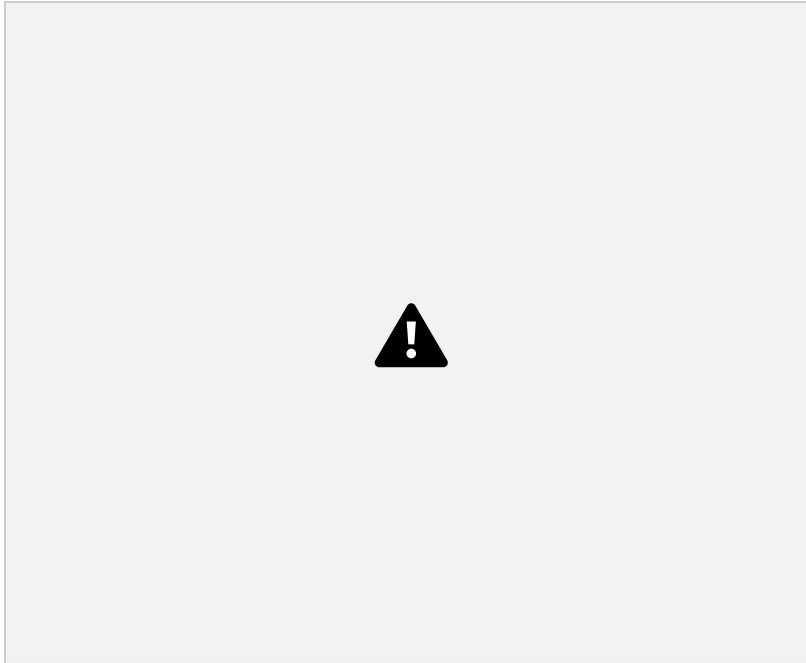
Selects documents where a field value is less than or equal to the specified



value.

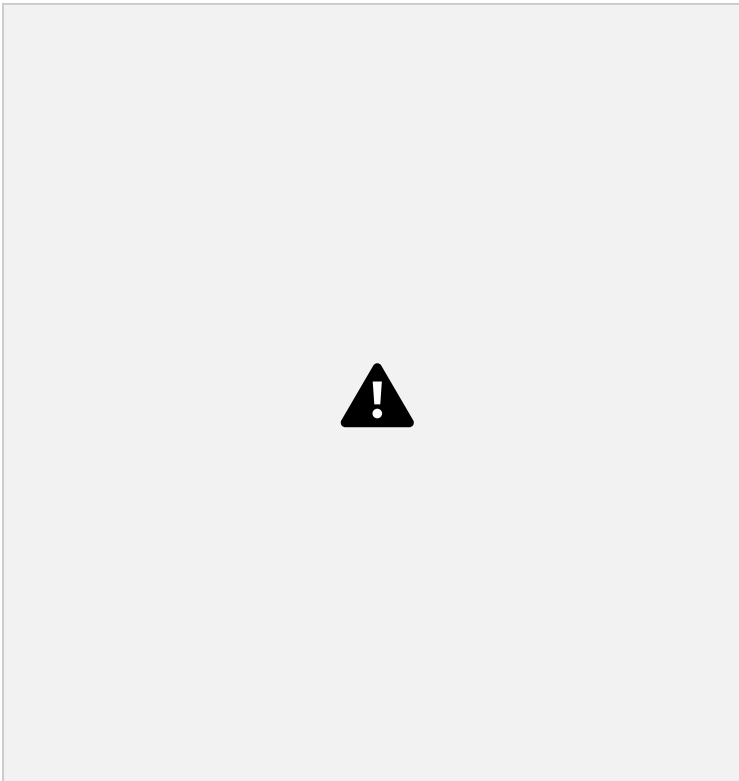
\$in

Selects documents where a field matches any value from the specified list.



\$nin

Selects documents where a field does not match any value from the specified



list.

6. Update Operations

updateOne()

Used to modify a single document that matches a specified condition.



`updateMany()`
Used to modify multiple documents that match a specified condition.



7. Delete Operations

`deleteOne()`

Used to delete a single document that matches a specified condition.



`deleteMany()`

Used to delete multiple documents that match a specified condition.



8. Aggregation Operations

`$match`

Used to filter documents based on specified conditions.



\$group and \$sum

\$group

Used to group documents based on a particular field and perform calculations on each group.

\$sum

Used to calculate the total of numeric values or count documents.



\$avg

Used to calculate the average value of a numeric field.



`$count`

Used to count the number of documents in the collection or result.



`$project`

Used to select, exclude, or create specific fields in the output.



`$sort`

Used to arrange documents in ascending or descending order.



`$unwind`

Used to convert each element of an array into a separate document.



\$lookup

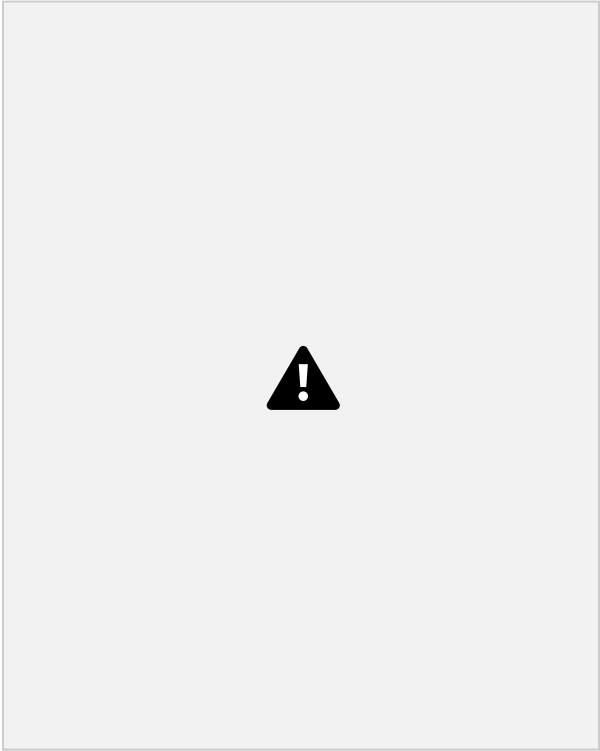
Used to combine related data from two different collections, similar to a JOIN in SQL.

Join Projects with Clients

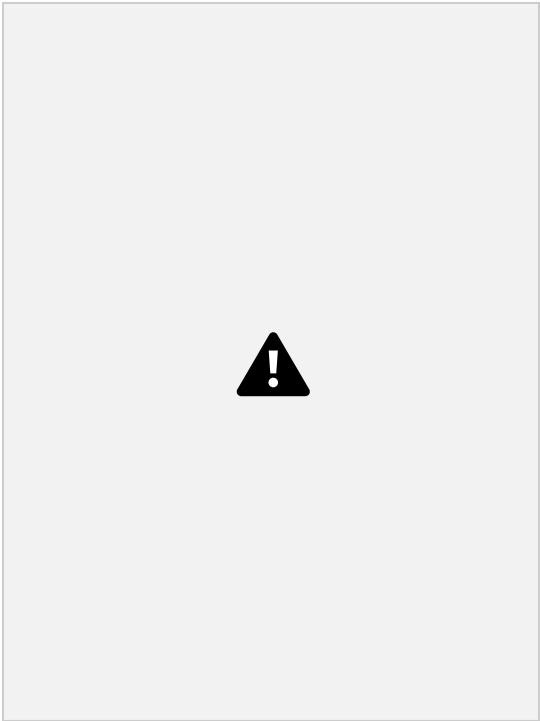




\$lookup with \$project



Group Projects by Category



Payment Aggregation



Projects by Client

